

PART-A.

3

1.1 (c) ✓

1.2 (a) ✓

1.3 (c) ✓

06

1.4 All are correct. ✓

$$1.5 \quad 3(u \cdot v) - 2(v \cdot v) + 3(u \cdot w) + 3(w \cdot w)$$

$$3(0) - 2(1) + 3(0) + 3(1)$$

4

∴ (a) ✓

PART-B.

1

$$\vec{u} \cdot \vec{v} = \vec{u} \cdot \vec{w}$$

$$\therefore |\vec{u}| |\vec{v}| \cos \theta_1 = |\vec{u}| |\vec{w}| \cos \theta_2$$

$$\cos \theta_1 |\vec{v}| = |\vec{w}| \cos \theta_2$$

where θ_1 is angle between u and v ,

θ_2 is angle between u and w .

Hence we can tell $|\vec{v}| \cos \theta_1 = |\vec{w}| \cos \theta_2$ and can comment on their magnitudes only.

No Inference can be draw on whether $\vec{v} = \vec{w}$

Hence, Reject.

2) If X is a non Zero eigenvector of A

$$AX = \lambda X$$

$$[A - \lambda I]X = 0$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 5-\lambda & 4 \\ 4 & 5-\lambda \end{vmatrix} = 0$$

$$(5-\lambda)^2 - 4^2 = 0$$

$$(5-\lambda-4)(5-\lambda+4) = 0$$

$$(1-\lambda)(9-\lambda) = 0 \Rightarrow \lambda = 1, 9$$

$$\lambda^2 - 10\lambda + 9$$

6

⇒ If $\lambda = 1$.

$$\begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

$$\equiv \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \theta$$

$$\begin{aligned} R_2 &\leftarrow R_2 - R_1 \\ R_1 &\leftarrow R_1 \div 4 \end{aligned}$$

$$x + y = 0.$$

$$x = -y$$

$$\therefore \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ y \end{bmatrix} = y \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

∴ For $\lambda = 1$ the eigen vectors are of format

$$k \begin{bmatrix} -1 \\ 1 \end{bmatrix} \text{ for } k \in \mathbb{R}.$$

⇒ If $\lambda = 9$

$$\begin{bmatrix} -4 & 4 \\ 4 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} R_2 &\leftarrow R_2 + R_1 \\ R_1 &\leftarrow R_1 \div 4 \end{aligned}$$

$$\equiv \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \theta$$

$$-x + y = 0$$

$$x = y$$

$$\therefore \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ y \end{bmatrix} = y \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

∴ For $\lambda = 9$ the eigen vectors of A are in format

$$k \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ for } k \in \mathbb{R}$$

$$(3) \quad W = \text{span}(v_1, v_2) = \text{span}\left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}\right)$$

$$W^\perp = \{ \vec{x} \mid \vec{x} \cdot v_1 = 0 \wedge \vec{x} \cdot v_2 = 0 \}$$

$$\text{Let } x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\vec{x} \cdot v_1 = 0 \quad \therefore \quad x_1(1) + x_2(1) + x_3(0) = 0$$

$$\vec{x} \cdot v_2 = 0 \quad \therefore \quad x_1(1) + x_2(0) + x_3(-1) = 0$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$Ax = 0$$

RRJ(A):

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & -1 \end{bmatrix} \quad R_2 \leftarrow R_1 - R_2$$

Let x_3 be free variable.

$$x_2 + x_3 = 0 \quad \Rightarrow \quad x_2 = -x_3$$

$$x_1 + x_2 = 0 \quad \Rightarrow \quad x_1 = -x_2 = x_3$$

$$\therefore X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_3 \\ -x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$\therefore$ Orthogonal
Complement = $\text{span}\left(\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}\right)$
Subspace of
 W

(4) Nullspace (A) = $\{x \mid Ax = 0\}$

$\Rightarrow$ RREF(A):

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

x_1 is free variable.

$$0(x_1) + 1(x_2) = 0$$

$$x_2 = 0$$

$$\therefore \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \end{bmatrix} \therefore \text{Nullsp}(A) = \text{span} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right)$$

$$\dim(\text{Nullsp}(A)) = 1$$

$\Rightarrow$ Colspace (A) = $\text{Span}(\text{column vectors of } A)$.

Linearly Independent.

$$\therefore \text{Colsp}(A) = \text{span} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) \because 0 \text{ vector is linearly dependent}$$

$$\dim(\text{Colsp}(A)) = 1$$

$\Rightarrow$ Observation:

$$\dim(N_A) + \dim(\text{Colsp}(A)) = \# \text{ of columns}$$

~~Usually~~ # free variables + # pivot elements = # of columns is true every time.

For a matrix # pivot elements = Rank of the matrix.

For a square matrix $\text{Rank}(A) = \text{Rank}(A^T)$.

{ Logic: Rank is the largest square matrix with non-zero determinant. A square matrix is already square, so transpose wouldn't affect it }.

Hence for a square matrix. # LI rows = # LI columns = Rank.

$$\dim(\text{Colsp}(A^T)) = \dim(\text{Colsp}(A)) = \text{Rank}$$

$$\text{WKT } |A^T| + |N_A| = \# \text{ columns} = n \leftarrow \text{for square matrix.}$$

$$|A| + |N_{A^T}| = \# \text{ rows} = n \leftarrow$$

Hence we can generalize.

$\dim(\text{Nullspace}(A)) = \dim(\text{Nullspace}(A^T))$ for square matrices

(5) $N_A = \{x \mid Ax = 0\}$.

$$\Rightarrow \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & -1 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_1 - R_2} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & -1 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_2 + R_3} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} = \text{RREF}(A).$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

$$x_1 + x_2 = 0.$$

$$0 + x_2 = 0.$$

$$\therefore x_1 = x_2 = 0.$$

$$N_A = \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\} \text{ basis of } N_A = \{ \}. \text{ Since zero vector has no basis.}$$

$\Rightarrow N(A^T) = \{x \mid A^T x = 0\}$.

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_1 - R_2} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \text{RREF}(A^T).$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_2 + x_3 = 0 \Rightarrow x_2 = -x_3 \text{ \{ Let } } x_3 = \text{free variable} \text{ \}}$$

$$x_1 + x_2 = 0 \Rightarrow x_1 = -x_2 = x_3$$

$$\therefore \begin{bmatrix} x_3 \\ -x_3 \\ x_3 \end{bmatrix} \quad N(A^T) = \text{span} \left(\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right).$$

(*)

(6)

$\Rightarrow \text{Colsp}(A)$
 $\text{span} \left(\underbrace{\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}}_{v_1}, \underbrace{\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}}_{v_2} \right)$

Both are linearly independent

$\therefore x_1 = x_2 = 0$ for
 $x_1 v_1 + x_2 v_2 = 0$

$\Rightarrow \text{Colsp}(A^T)$:

$\text{span} \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \end{bmatrix} \right)$

But it can be further reduced.

To find if $x_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ only for $x_1 = x_2 = x_3 = 0$.

$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \equiv \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & -1 \end{bmatrix}$

$\left. \begin{array}{l} x_1 + x_2 = 0 \\ x_2 + x_3 = 0 \end{array} \right\} \text{ Infinite solutions.}$

pivot variables.

$\therefore \text{Col}(A^T) = \text{span} \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right)$

Table:

Subspace	Basis	Dimension
$N(A)$	$\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \}$	0
$\text{Colsp}(A)$	$\{ [1 \ 1 \ 0]^T, [1 \ 0 \ -1]^T \}$	2
$N(A^T)$	$\{ [1 \ -1 \ 1]^T \}$	1
$\text{Colsp}(A^T)$	$\{ [1 \ 1]^T, [1 \ 0]^T \}$	2

$0 + 2 = n = 2$

$2 + 1 = m = 3$

12 ✓

(6) 2 vectors a, b are orthonormal if.

$$a \cdot b = 0, \text{ and}$$

$$a \cdot a = b \cdot b = 1.$$

$$\phi_1 \cdot \phi_2 = 1(1) + 1(-1) + 1(1) + 1(-1) = 0.$$

$$\phi_1 \cdot \phi_3 = 1(1) + 1(1) + 1(-1) + 1(-1) = 0.$$

$$\phi_1 \cdot \phi_4 = 1(1) + 1(-1) + 1(-1) + 1(1) = 0.$$

$$\phi_2 \cdot \phi_3 = 1(1) - 1(1) + 1(-1) - 1(-1) = 0.$$

$$\phi_2 \cdot \phi_4 = 1(1) - 1(-1) + 1(-1) - 1(1) = 0.$$

$$\phi_3 \cdot \phi_4 = 1(1) + 1(-1) + 1(-1) - 1(1) = 0.$$

$\therefore \phi_1, \phi_2, \phi_3, \phi_4$ are orthogonal, but not orthonormal since $\phi_i \cdot \phi_i = 1^2 + 1^2 + 1^2 + 1^2 = 4$.

To make it orthonormal divide each component by $\sqrt{4} = 2$.

$$\text{Let } u = x_1 \phi_1 + x_2 \phi_2 + x_3 \phi_3 + x_4 \phi_4.$$

$$x_i = \frac{u \cdot \phi_i}{\phi_i \cdot \phi_i}$$

$$\therefore x_1 = \frac{u \cdot \phi_1}{\phi_1 \cdot \phi_1} = \frac{1}{4} (1+2+3+4) = 5/2$$

$$x_2 = \frac{u \cdot \phi_2}{\phi_2 \cdot \phi_2} = \frac{1}{4} (1-2+3-4) = -1/2$$

$$x_3 = \frac{u \cdot \phi_3}{\phi_3 \cdot \phi_3} = \frac{1}{4} (1+2-3-4) = -2/2 = -1$$

$$x_4 = \frac{u \cdot \phi_4}{\phi_4 \cdot \phi_4} = \frac{1}{4} (1-2-3+4) = 0.$$

$$\therefore u = \frac{5}{2} \phi_1 - \frac{1}{2} \phi_2 - \phi_3.$$

12

$$\begin{matrix} 0 & 1 \\ 0 & 0 \end{matrix}$$

$$\dim(\text{NA}) + \dim(\text{colsp}(A)) = n = 2.$$

But NA .

$$\text{No. of FU} + \text{No. of PE} = n.$$

$$\begin{bmatrix} 3 & 2 & 1 \\ 6 & 4 & 2 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 2 & 1 \\ 0 & 0 & 0 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{3} & \frac{1}{2} & \frac{2}{1} \\ 0 & \frac{1}{0} & \frac{2}{0} \\ 0 & 0 & 0 \end{bmatrix}$$

$$\dim \text{colsp} = 2 \quad |C| = 2$$

$$|NA| = 1$$

$$\begin{bmatrix} 3 & 6 & 1 \\ 2 & 4 & 1 \\ 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 1 \\ 3 & 6 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} \frac{1}{3} & \frac{2}{2} & \frac{2}{2} \\ 0 & 0 & \frac{1}{3} \\ 0 & 0 & \frac{1}{5} \end{bmatrix}$$

$$\dim |CAT| = 2$$

$$|NA| = 1$$

$$|CAT| + |NA| = n.$$